SECTION 2 · SUBSECTION 2.1
COMPUTING EQUIPMENT AND SOFTWARE

2.1.1 INTRODUCTION.

The backend is a Python 3 application with no framework beyond FastAPI at its edge. Analysis depends on five libraries that do the substantive work: tree-sitter for syntax, NetworkX for graph algorithms, Radon for Python complexity, ChromaDB for vector storage, and the OpenAI client pointed at NVIDIA NIM. The one idea the reader must leave with is that *requirements.txt* declares floors only; the versions in this handbook are the ones resolved in the local virtual environment, not a locked contract (figure 2.1-1).

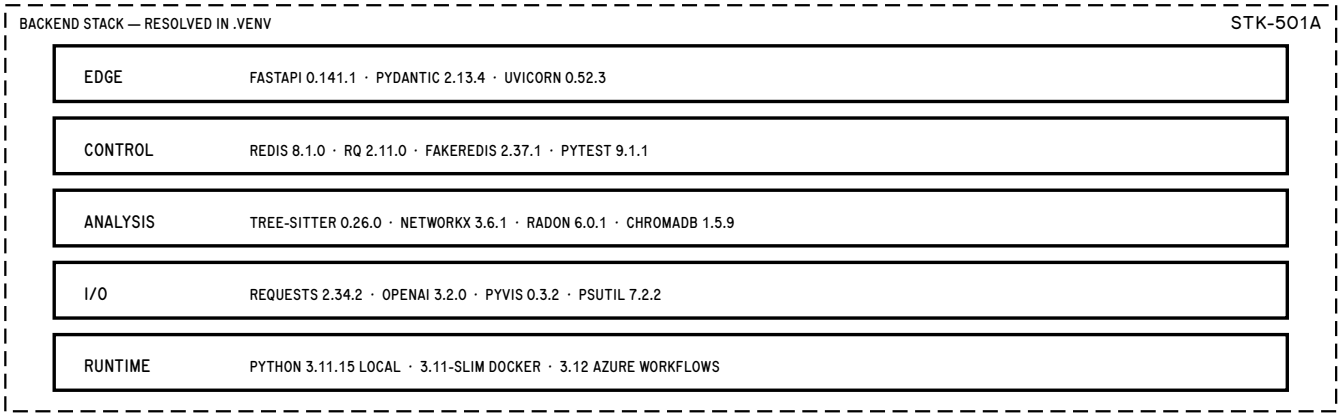


Figure 2.1-1. Backend Dependency Layers and Resolved Versions

2.1.2 FUNCTIONAL DESCRIPTION.

FastAPI parses and validates requests with Pydantic models, exposes the routes, and delegates every long operation to a job runner. Requests and the OpenAI client carry out HTTPS calls to GitHub and NVIDIA. Tree-sitter supplies a concrete syntax tree per language; the parsers in *src/parsers/* convert those trees to symbol tables. NetworkX holds the file graph in memory and supplies cycle detection, topological sorting and PageRank. Radon computes Python cyclomatic complexity; the JavaScript and generic paths use a custom branch counter. ChromaDB persists chunk embeddings under *./chroma_db*, and RQ moves job identifiers through Redis.

2.1.2.1 SQLite Substitution.

At the top of *src/api/main.py*, before any ChromaDB import, the module imports *pysqlite3* and replaces *sys.modules["sqlite3"]* with it. The comment states that Azure App Service ships SQLite 3.31 while ChromaDB requires at least 3.35. The substitution is unconditional, so it also runs locally; it is an environment workaround, not an analysis feature.

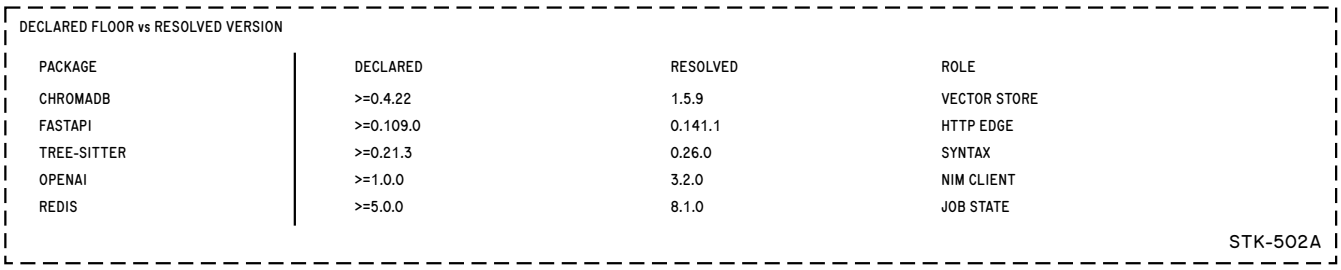


Figure 2.1-2. Declaration Floors and Local Resolution

2.1.3 PERFORMANCE AND DESIGN DATA.

Table 2.1-1. Backend Software and Actual Role

Package	Resolved	Role in this system
chromadb	1.5.9	Per-job collection of embedded code chunks
networkx	3.6.1	DiGraph, simple_cycles, topological_sort, pagerank
radon	6.0.1	Python cyclomatic complexity and line metrics
tree-sitter	0.26.0	Parser runtime for eight language grammars
fastapi	0.141.1	Routes, validation, middleware
pydantic	2.13.4	Request and response schemas
openai	3.2.0	OpenAI-compatible client for NVIDIA NIM
redis / rq	8.1.0 / 2.11.0	Job records and distributed queue
pytest	9.1.1	476-test verification suite

CAUTION

There is no Python lockfile. Every requirement is a lower bound, so two hosts can resolve different versions. The frontend does have a lockfile. Reproduce the versions in table 2.1-1 before attributing a defect to the code.

```
STACK CHECK                          MAINTAINER
1 READ - requirements.txt floors
2 RUN - python -m pip list
  √ MATCH - table 2.1-1
3 CHECK - Python 3.11 or 3.12
4 VERIFY - sqlite3 module is pysqlite3
```